SIT723 Research Techniques & Applications
Task 3.2P: Summarise Research Methods

This Task summarises the research methods used in the 40 papers reviewed as part of Task 2.1P. Papers are grouped by the research question they most directly support, followed by supporting methods and ethical considerations.

1. Datasets		
Paper	**Summary**	**Methodology**
S. Mohan and D. Li, 2019, "MedMentions: A large biomedical corpus annotated with UMLS concepts," Proceedings of the 2019 Conference on Automated Knowledge Base Construction (AKBC), DOI / Identifier: arXiv:1902.09476	Introduces MedMentions, one of the largest biomedical entity-linking datasets, with 4,392 PubMed abstracts and 350,000+ UMLS-linked mentions. The ST21pv subset restricts to 21 preferred semantic types, providing CUI-level ground truth. This is the primary dataset in the present study (n = 550 stratified sample).	• 4,392 PubMed abstracts collected and expert annotated. • 350,000+ entity mentions linked to UMLS CUIs. • ST21pv: 21 preferred semantic type categories. • 97.3% annotation precision confirmed by two independent reviewers. • Publicly released under CC0 licence.
P. Rajpurkar, R. Jia, and P. Liang, 2018, "Know what you don't know: Unanswerable questions for SQuAD," Proceedings of the 56th Annual Meeting of the ACL, DOI: 10.18653/v1/P18-2124	Introduces SQuAD 2.0 combining 100,000 answerable and over 50,000 unanswerable adversarially designed questions. Used in the present study as the general-domain cross-domain validation dataset (n = 200 answerable questions stratified by answer type).	• 100,000 answerable + over 50,000 unanswerable questions. • Unanswerable questions adversarially designed to resemble answerable ones. • Human annotation via crowdsourcing with quality checks. • Requires models to output a span or abstain.
G. Tsatsaronis, G. Balikas, P. Malakasiotis, I. Partalas, M. Zschunke, M. R. Alvers, D. Weissenborn, A. Krithara, S. Petridis, D. Polychronopoulos, Y. Almirantis, J. Pavlopoulos, N. Baskiotis, P. Gallinari, T. Artiéres, A.-C. Ngonga	Biomedical factoid and yes/no QA challenge linked to PubMed literature. Used as an intermediate biomedical validation set (n = 150) enabling	• Annual biomedical QA shared task linked to PubMed. • Factoid (~90) and yes/no (~60) question types. • Years 2021-2023 selected for

<table>
<tr><td>Ngomo, N. Heino, E. Gaussier, L. Barrio-Alvers, M. Schroeder, I. Androutsopoulos, and G. Paliouras, 2015, “An overview of the BIOASQ large-scale biomedical semantic indexing and question answering competition,” BMC Bioinformatics, DOI: 10.1186/s12859-015-0564-6.</td><td>the domain-continuum analysis: MedMentions (clinical NER) to BioASQ (biomedical QA) to SQuAD 2.0 (general QA).</td><td>contemporary language coverage.
• Publicly available at bioasq.org for registered participants.</td></tr>
<tr><td colspan="3">2. Research Methods for RQ1: Characterisation of Semantic Instability</td></tr>
<tr><td colspan="3">Which linguistic properties of meaning-preserving perturbations are the strongest independent predictors of semantic entropy in clinical LLM outputs, and do these predictors interact?</td></tr>
<tr><td>Paper</td><td>Summary</td><td>Methodology</td></tr>
<tr><td>L. Kuhn, Y. Gal, and S. Farquhar, 2023, “Semantic uncertainty: Linguistic invariances for uncertainty estimation in natural language generation,” 11th International Conference on Learning Representations (ICLR), DOI / Identifier: openreview.net/forum?id=VD-AYtP0dve</td><td>Introduces semantic entropy: groups NLG outputs by shared meaning using bidirectional NLI and computes Shannon entropy over meaning-level clusters. Core theoretical and methodological foundation for the entropy computation in the present study.</td><td>• Sampled multiple outputs per input from generative LLMs.
• Clustered outputs using bidirectional NLI for semantic equivalence.
• Computed $H(x0) = -sum\ p(si)\ log2\ p(si)$ over clusters.
• Compared against token-level baselines (predictive entropy, p(True)).
• Evaluated on TriviaQA and CoQA. Model-agnostic.</td></tr>
<tr><td>S. Farquhar, J. Kossen, L. Kuhn, and Y. Gal, 2024, “Detecting hallucinations in large language models using semantic entropy,” Nature, DOI: 10.1038/s41586-024-07421-0</td><td>Extends semantic entropy to hallucination detection. Shows semantically diverse outputs correlate with factual errors without requiring labelled hallucination data. Directly validates the core metric used in the present study.</td><td>• Sampled multiple outputs per prompt from generative LLMs.
• Bidirectional entailment clustering for meaning equivalence.
• Computed entropy over meaning-level cluster distribution.
• Zero-resource: no labelled hallucination data required.
• Published in Nature.</td></tr>
<tr><td>M. Sclar, Y. Choi, Y. Tsvetkov, and A. Suhr, 2024, “Quantifying language models’ sensitivity to spurious</td><td>Demonstrates up to 76 percentage point performance swings</td><td></td></tr>
</table>

features in prompt design or: How I learned to start worrying about prompt formatting," 12th International Conference on Learning Representations (ICLR), DOI / Identifier: arXiv:2310.11324	from meaning-preserving formatting changes. Shows benchmark accuracy can be misleading under surface variation, directly motivating why semantic stability must be measured explicitly.	• Applied to multiple LLMs across tasks (GPT-3.5, LLaMA). • Reported up to 76-point accuracy variance from formatting alone. • No fine-tuning required.
J. Zhuo, S. Zhang, X. Fang, H. Duan, D. Lin, and K. Chen, 2024, "ProSA: Assessing and understanding the prompt sensitivity of LLMs," Findings of the Association for Computational Linguistics: EMNLP 2024, DOI / Identifier: 10.18653/v1/2024.findings-emnlp.108	Introduces PromptSensiScore, a systematic metric for prompt sensitivity. Shows sensitivity varies across tasks and that few-shot examples can partially reduce the problem. Connects prompt robustness to decoding confidence.	• Defined PromptSensiScore across prompt paraphrase sets. • Evaluated across multiple datasets showing task-dependent variation. • Tested few-shot prompting as a sensitivity reduction strategy. • Linked prompt robustness to model decoding confidence.
H. Raj, V. Gupta, D. Rosati, and S. Majumdar, 2023, "Semantic consistency for assuring reliability of large language models," arXiv preprint, DOI / Identifier: arXiv:2308.09138.	Shows LLMs frequently produce different answers to semantically equivalent questions. Demonstrates the gap between one-time correctness and stable understanding. Introduces Ask-to-Choose as a consistency improvement method.	• Presented semantically equivalent paraphrase sets to LLMs. • Measured consistency without gold labels or supervised retraining. • Proposed Ask-to-Choose prompting to improve consistency. • Evaluated across multiple models and tasks.
X. Wang, J. Wei, D. Schuurmans, Q. V. Le, E. Chi, S. Narang, A. Chowdhery, and D. Zhou, 2023, "Self-consistency improves chain of thought reasoning in language models," 11th International Conference on Learning Representations (ICLR), DOI / Identifier:10.48550/arXiv.2203.11171	Introduces self-consistency decoding: generates multiple reasoning paths and selects by majority agreement. Shows output agreement across samples is linked to reliability.	• Generated multiple chain-of-thought reasoning paths per question. • Majority voting over final answers across sampled paths.

	Conceptually related to semantic entropy both rely on stability across multiple outputs.	• Evaluated on arithmetic and commonsense reasoning benchmarks. • No model fine-tuning required.
G. G. Şahin, 2022, “To augment or not to augment? A comparative study on text augmentation techniques for low-resource NLP,” Computational Linguistics, DOI: 10.1162/coli_a_00425.	Compares back-translation, synonym replacement, and syntactic reordering across languages and tasks. Finds no universally superior strategy. Directly informs the multi-method perturbation design of the present study (k = 8 perturbations per instance across four methods).	• Compared back-translation, synonym replacement, syntactic reordering. • Evaluated across multiple languages and NLP task types. • Assessed augmentation benefit as a function of dataset size. • Found task- and language-specific benefits with no universal winner.
P. Manakul, A. L. Liusie, and M. J. F. Gales, 2023, “SelfCheckGPT: Zero-resource black-box hallucination detection for generative large language models,” Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP), DOI: 10.18653/v1/2023.emnlp-main.557	Introduces SelfCheckGPT: checks consistency across multiple sampled outputs using NLI, BERTScore, or n-gram overlap without clustering or labelled data. Serves as the primary methodological comparison baseline for semantic entropy in the present study.	• Sampled multiple responses per prompt from the same LLM. • Checked consistency using NLI, BERTScore, and n-gram signals. • Zero-resource: no knowledge base or labelled data required. • Key distinction from semantic entropy: no formal meaning clustering.
J. Novikova, C. Anderson, B. Blili-Hamelin, D. Rosati, and S. Majumdar, 2025, “Consistency in language models: Current landscape, challenges, and future directions,” arXiv preprint, DOI / Identifier: arXiv:2505.00268	Reviews multiple dimensions of consistency in LLMs: factual, hypothetical, compositional, and moral. Shows models perform especially poorly on hypothetical and compositional consistency. Positions semantic stability under paraphrase within the	• Reviewed factual, hypothetical, compositional, and moral consistency. • Found major gaps in hypothetical and compositional consistency. • Most research concentrated on English text-to-text models.

	wider consistency literature.	• Helps position the present study in the consistency literature.
3. Research Methods for RQ2: Accuracy-Stability Dissociation		
Does semantic entropy detect instability in high-accuracy LLM outputs that traditional evaluation metrics fail to reveal, and does this dissociation hold across encoder-only and generative architectures?		
Paper	**Summary**	**Methodology**
P. Hager, F. Jungmann, R. Holland, K. Bhatt, E. Butcher, P. Bhatt, and D. Rückert, 2024, "Evaluation and mitigation of the limitations of large language models in clinical decision-making," Nature Medicine, DOI: 10.1038/s41591-024-03097-1	Tests LLMs on realistic clinical tasks, finding they perform much worse than benchmarks suggest. Models struggle with guideline adherence, lab interpretation, and clinical reasoning. Primary evidence for the evaluation illusion in medicine.	• Tested LLMs on physician-validated realistic clinical scenarios. • Tasks: guideline adherence, lab interpretation, clinical reasoning. • Compared benchmark accuracy against real clinical task performance. • Found large gaps between benchmark and clinical task performance.
M. Agrawal, I. Y. Chen, F. Gulamali, and S. Joshi, 2025, "The evaluation illusion of large language models in medicine," npj Digital Medicine, DOI: 10.1038/s41746-025-01963-x.	Argues apparent medical LLM success is exaggerated by benchmark design and metric choice. Automated scores misalign with expert clinical judgement. Directly supports the need for stronger evaluation such as semantic entropy, framing Gap 4 in the present study.	• Systematic critique of medical LLM benchmark evaluation practices. • Reviewed mismatch between automated metrics and expert judgement. • Analysed benchmark design choices that inflate apparent performance.
S. Zhou, Y. Cheng, W. Zhang, X. Liang, C. Li, Y. Li, Y. Hu, and J. Liu, 2025, "Uncertainty-aware large language models for explainable disease diagnosis," npj Digital Medicine, DOI: 10.1038/s41746-025-02071-6.	Introduces ConfiDx, an uncertainty-aware diagnosis framework with explicit uncertainty labels during fine-tuning. Shows uncertainty	• Fine-tuned LLMs with explicit uncertainty labels (ConfiDx framework).

	modelling reduces misdiagnosis. Supervised approach complementing the present study's unsupervised semantic entropy metric.	• Evaluated on clinical diagnosis tasks with ambiguous evidence. • Showed reduced misdiagnosis rates with uncertainty-aware training. • Supervised approach contrasts with unsupervised semantic entropy.
S. Kadavath, T. Conerly, A. Askell, T. Henighan, D. Drain, E. Perez, N. Schiefer, Z. Hatfield-Dodds, N. DasSarma, E. Tran-Johnson et al., 2022, “Language models (mostly) know what they know,” arXiv, DOI / Identifier: arXiv:2207.05221	Examines whether LLMs can estimate their own correctness using p(True). Models appear calibrated on average but are unreliable at the individual instance level. Supports the need for external metrics such as semantic entropy over self-reported confidence.	• Queried Anthropic language models (800M–52B parameters) with p(True) self-assessment prompts. • Compared model-reported confidence against empirical accuracy using calibration charts. • Evaluated calibration (ECE) and discriminative power (AUROC) across model sizes and tasks. • Found models are well-calibrated on average but self-evaluation remains imperfect at the individual sample level.
C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, 2017, “On calibration of modern neural networks,” Proceedings of the 34th International Conference on Machine Learning (ICML), DOI / Identifier: PMLR: proceedings.mlr.press/v70/guo17a	Foundational calibration study showing neural networks are often overconfident even when wrong. Proposes temperature scaling as post-hoc calibration. Helps separate confidence calibration from semantic consistency in the present study.	• Computed Expected Calibration Error (ECE) across multiple architectures. • Used reliability diagrams to visualise confidence vs. accuracy gaps. • Proposed temperature scaling as a simple post-hoc calibration fix. • Established distinction between accuracy, confidence, and calibration.

G. G. Şahin, 2022, "To augment or not to augment? A comparative study on text augmentation techniques for low-resource NLP," Computational Linguistics, DOI: 10.1162/coli_a_00425.	Compares multiple text augmentation strategies for low-resource NLP and shows that augmentation effectiveness depends on the task, language family, and model type. Useful for motivating perturbation design and for justifying why augmentation methods should be evaluated carefully rather than assumed to help uniformly.	• Systematically compares syntax-, token-, and character-level augmentation methods for low-resource NLP. • Evaluates augmentation on part-of-speech tagging, dependency parsing, and semantic role labeling across diverse languages. • Finds augmentation is often more helpful for dependency parsing than for the other tasks. • Reports that character-level methods are the most consistent, while synonym replacement and some syntactic methods give mixed results.
T. Savage, J. Wang, R. Gallo, A. Boukil, V. Patel, S. A. A. Safavi-Naini, A. Soroush, and J. H. Chen, 2024, "Large language model uncertainty proxies: Discrimination and calibration for medical diagnosis and treatment," Journal of the American Medical Informatics Association, DOI: 10.1093/jamia/ocae254.	Evaluates output similarity and embedding-based consistency as uncertainty proxies in medical LLM tasks. Consistency-style signals can help distinguish correct from incorrect answers. Provides a direct medical-domain comparison point for the entropy-based approach.	• Applied output similarity and embedding consistency measures. • Evaluated discrimination and calibration in medical diagnosis tasks. • Compared multiple consistency signal types in a medical context. • Shows consistency-based uncertainty is useful but imperfect in medicine.
Z. Ji, N. Lee, R. Frieske, T. Yu, D. Su, Y. Xu, E. Ishii, Y. J. Bang, D. Chen, W. Dai, A. Madotto, and P. Fung, 2023, "Survey of hallucination in natural language generation," ACM Computing Surveys, DOI: 10.1145/3571730.	Broad overview of hallucination in NLG distinguishing intrinsic (contradiction) from extrinsic (fabrication) hallucinations. Reviews causes at data, model design, and inference levels. Explains why semantic instability	• Systematic review of hallucination types across NLG systems. • Taxonomy: intrinsic (contradiction) vs. extrinsic (fabrication). • Reviewed root causes at data, model, and inference levels.

	matters unstable outputs are linked to hallucinated responses.	• Assessed detection and mitigation strategies across tasks.
L. Huang, W. Yu, W. Ma, W. Zhong, Z. Feng, H. Wang, Q. Chen, W. Peng, X. Feng, B. Qin, and T. Liu, 2025, "A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions," ACM Transactions on Information Systems, DOI: 10.1145/3703155.	Updates hallucination research for the LLM era, separating factuality from faithfulness hallucinations. Reinforces that reliable behaviour should include consistency across semantically equivalent inputs, not just high one-shot accuracy.	• Extended hallucination taxonomy for instruction-following LLMs. • Separated factuality hallucinations from faithfulness hallucinations. • Reviewed retrieval, factchecking, and reinforcement mitigation methods. • Positioned semantic consistency as a key evaluation requirement.
K. Singhal, S. Azizi, T. Tu, S. S. Mahdavi, J. Wei, H. W. Chung, N. Scales et al., 2023, "Large language models encode clinical knowledge," Nature, DOI: 10.1038/s41586-023-06291-2	Presents Med-PaLM showing LLMs can approach expert-level on medical QA benchmarks. Simultaneously highlights ongoing issues with uncertainty communication, harm, and bias. Shows high benchmark accuracy does not mean safe or stable understanding.	• Evaluated Med-PaLM on USMLE-style medical licensing exam questions. • Collected physician and non-physician baselines for comparison. • Multi-shot prompting with curated medical examples. • Assessed safety, harm, bias, and uncertainty separately from accuracy.
4. Research Methods for RQ3: Domain Adaptation and Semantic Stability		
Does domain-specific biomedical pretraining reduce semantic entropy at equivalent scale, and does this advantage generalise from concept normalisation to general-domain QA?		
Paper	**Summary**	**Methodology**
J. Lee, W. Yoon, S. Kim, D. Kim, S. Kim, C. H. So, and J. Kang, 2020, "BioBERT: A pre-trained biomedical language representation model for biomedical text mining," Bioinformatics, DOI: 10.1093/bioinformatics/btz682	Foundational biomedical NLP model pretrained on PubMed and PMC data. Improves NER, relation extraction, and QA. In the present study, BioBERT (110M	• Continued pretraining from BERT-based on 4.5B words from PubMed + PMC. • Fine-tuned on biomedical NER, relation extraction, and QA.

	parameters) is the primary domain-specific encoder-only model in the within-scale comparison against BERT-base for RQ3.	• Compared against BERT-base and domain-specific baselines. • Showed consistent gains across all biomedical downstream tasks.
Y. Gu, R. Tinn, H. Cheng, M. Lucas, N. Usuyama, X. Liu, T. Naumann, J. Gao, and H. Poon, 2021, “Domain-specific language model pretraining for biomedical natural language processing,” ACM Transactions on Computing for Healthcare, DOI: 10.1145/3458754	Introduces PubMedBERT trained from scratch on purely biomedical text, outperforming mixed-domain alternatives. In the present study, PubMedBERT is an additional encoder-only model in the RQ3 comparison providing a second data point for the domain-adaptation hypothesis.	• Pretrained from random initialisation using PubMed abstracts only. • No general-domain text fully in-domain vocabulary and statistics. • Evaluated on 13 tasks in the BLURB biomedical NLP benchmark. • Achieved top BLURB results demonstrating value of in-domain pretraining.
Y. Labrak, A. Bazoge, E. Morin, P.-A. Gourraud, M. Rouvier, and R. Dufour, 2024, “BioMistral: A collection of open-source pretrained large language models for medical domains,” Findings of the Association for Computational Linguistics: ACL 2024, DOI / Identifier: 10.18653/v1/2024.findings-acl.348	Introduces BioMistral-7B built from Mistral-7B and adapted on biomedical corpora. Open-source models approach GPT-3.5 performance on medical QA. In the present study, BioMistral-7B (INT8, ~8 GB VRAM) is the primary domain-specific large model in the within-scale comparison against FLAN-T5-XXL.	• Fine-tuned Mistral-7B on biomedical corpora including PubMed and MedQA. • Evaluated on multiple medical QA benchmarks. • Compared against GPT-3.5, LLaMA-based, and other medical LLMs. • Released as open source under permissive licence.
R. Tinn, H. Cheng, Y. Gu, N. Usuyama, X. Liu, T. Naumann, J. Gao, and H. Poon, 2023, “Fine-tuning large neural language models for biomedical natural language processing,” Patterns (Cell Press), DOI: 10.1016/j.patter.2023.100729.	Studies fine-tuning stability in biomedical NLP. Shows PubMedBERT is more stable across runs than other BERT-style models. Directly relevant to RQ3 suggests domain	• Evaluated fine-tuning variance across random seeds and hyperparameters. • Compared PubMedBERT, BioBERT, and general-domain BERT variants.

	adaptation may affect output consistency beyond accuracy.	• Assessed stability across biomedical NLP tasks. • PubMedBERT showed lower variance across runs than alternatives.
R. Moradi and N. Samwald, 2022, “Improving the robustness and accuracy of biomedical language models through adversarial training,” Journal of Biomedical Informatics, DOI: 10.1016/j.jbi.2022.104114	Applies adversarial training in biomedical NLP and shows robustness and task accuracy can improve together. Shows stability is not only measurable but can also be improved during training. Provides a relevant biomedical comparison point for the model comparison in RQ3.	• Applied adversarial training examples during biomedical model fine-tuning. • Evaluated robustness and accuracy jointly on biomedical benchmarks. • Showed simultaneous improvement in both robustness and task performance. • Demonstrated adversarial training is viable in specialised domains.
5. Supporting Methods: UMLS Grounding, Adversarial Background, Clinical Context, and QA		
Paper	**Summary**	**Methodology**
M. Afshar, Y. Gao, D. Gupta, E. Croxford, and D. Demner-Fushman, 2024, “On the role of the UMLS in supporting diagnosis generation proposed by large language models,” Journal of Biomedical Informatics, DOI: 10.1016/j.jbi.2024.104707.	Finds UMLS-based metrics align better with expert judgement than ROUGE. Directly supports using UMLS CUI identifiers as a formal semantic equivalence criterion in the present study's entropy computation, superior to surface string matching.	• UMLS knowledge paths used to augment LLM diagnosis generation. • Compared UMLS-grounded metrics against ROUGE and BLEU. • Expert judgement ratings used as ground truth for metric alignment. • UMLS-grounded evaluation aligned more closely with expert scores.
N. J. Dobbins, 2025, “Generalizable and scalable multistage biomedical concept normalization leveraging large language models,” Research Synthesis Methods, DOI / Identifier: 10.1017/rsm.2025.9 / PMC12527512.	Shows LLMs perform poorly when directly mapping phrases to UMLS concepts but improve with structured pipelines. Demonstrates that variability in CUI assignment is real and	• Evaluated LLMs on direct UMLS concept normalisation (CUI assignment). • Compared direct LLM mapping against multi-stage pipeline approaches.

	measurable exactly what concept-level semantic entropy captures. Strongest motivation for Gap 2 in the present study.	• Used MedMentions and biomedical NER datasets for evaluation. • Demonstrated measurable variability in CUI assignment across phrasings.
O. Bodenreider, 2004, "The Unified Medical Language System (UMLS): Integrating biomedical terminology," Nucleic Acids Research, DOI: 10.1093/nar/gkh061	Foundational paper explaining the UMLS Metathesaurus architecture using CUIs to integrate 100+ biomedical vocabularies. UMLS CUIs are the formal basis for semantic equivalence in the present study's entropy computation the primary methodological innovation distinguishing it from prior semantic entropy work.	• Describes UMLS Metathesaurus architecture and CUI assignment methodology. • Covers semantic type hierarchy (134 types, 54 relations). • Describes MetaMap, MetamorphoSys, lvg, and related UMLS tools. • Formal basis for CUI-level semantic equivalence in the present study.
M. Neumann, D. King, I. Beltagy, and W. Ammar, 2019, "ScispaCy: Fast and robust models for biomedical natural language processing," Proceedings of the 18th BioNLP Workshop and Shared Task, ACL, DOI / Identifier: 10.18653/v1/W19-5034	Presents scispaCy, a biomedical NLP toolkit with entity recognition and UMLS linking. Primary implementation tool in the present study for mapping model outputs to UMLS CUIs for entropy computation. Efficient enough for large-scale pipeline use.	• Built on spaCy with biomedical entity models and UMLS linker. • UMLS entity linking with confidence scoring (0-1 scale). • Supports abbreviation detection for clinical text. • Widely validated in biomedical NLP pipeline benchmarks.
D. Jin, Z. Jin, J. T. Zhou, and P. Szolovits, 2020, "Is BERT really robust? A strong baseline for natural language attack on text classification and entailment," Proceedings of the AAAI Conference on Artificial Intelligence, DOI: 10.1609/aaai.v34i05.6311	Introduces TextFooler, a synonym-substitution adversarial attack preserving human-judged meaning while flipping model predictions. Shows strong NLP models are fragile under small	• Used synonym substitution constrained by POS tag and semantic similarity. • Attacked BERT and other NLP models on classification and NLI tasks.

	wording changes. Directly motivates using meaning-preserving perturbations to probe interpretation stability.	• Preserved human-judged semantic meaning while flipping predictions. • Demonstrated high model fragility under synonym-level changes.
J. Ebrahimi, A. Rao, D. Lowd, and D. Dou, 2018, "HotFlip: White-box adversarial examples for text classification," Proceedings of the 56th Annual Meeting of the ACL (Short Papers), DOI: 10.18653/v1/P18-2006	Introduces gradient-based adversarial attacks at character or token level. Not fully meaning preserving but establishes the fragility of text models under minimal perturbation. Important background for the adversarial robustness literature informing the perturbation design of the present study.	• Used model gradients to find minimal token or character edits. • Applied to text classification models (CNN, LSTM). • Showed large behavioural shifts from very small input changes. • Demonstrated vulnerability of text models to atomic perturbations.
S. Goyal, S. Doddapaneni, M. M. Khapra, and B. Ravindran, 2023, "A survey of adversarial defenses and robustness in NLP," ACM Computing Surveys, DOI: 10.1145/3593042	Reviews adversarial attacks and defenses, distinguishing meaning-preserving from meaning-changing perturbations. An attack can succeed while still changing meaning this distinction is critical for the present study which uses only meaning-preserving perturbations verified through a six-gate quality pipeline.	• Comprehensive survey of NLP adversarial attacks and defences. • Distinguished semantic preservation from attack success rate. • Reviewed character-level, word-level, and sentence-level perturbations. • Justified restricting the present study to meaning-preserving perturbations.
X. Xu, K. Kong, N. Liu, L. Cui, D. Wang, J. Zhang, and M. Kankanhalli, 2024, "An LLM can fool itself: A prompt-based adversarial attack," 12th International Conference on Learning Representations (ICLR), DOI / Identifier: OpenReview: VVgGbB9TNV.	Presents PromptAttack where an adversarial LLM generates perturbations that trick another LLM. Shows that prompt sensitivity reflects deeper representational instability rather than only surface formatting issues. Supports	• Used an adversarial LLM to generate prompt perturbations. • Target model manipulated through prompt design alone. • Measured attack success rates across benchmark tasks.

	perturbation-based evaluation as a probe of representational robustness.	• Demonstrated instability is partly intrinsic to model representations.
S. S. Alahmari, L. Hall, P. R. Mouton, and D. Goldgof, 2025, "Large language models robustness against perturbation," Scientific Reports, DOI: 10.1038/s41598-025-29770-0	Evaluates contemporary LLMs under perturbations including keyboard noise and word replacement. Larger models are more robust, but instability persists. Supports that scale alone does not solve semantic stability it must be measured explicitly rather than assumed to improve automatically.	• Evaluated multiple LLMs under keyboard noise and word-level perturbations. • Compared robustness across model families and parameter scales. • Found larger models more robust but still imperfect. • Showed semantic instability persists in state-of-the-art systems.
J. Wei, X. Wang, D. Schuurmans, M. Bosma, B. Ichter, F. Xia, E. Chi, Q. V. Le, and D. Zhou, 2022, "Chain-of-thought prompting elicits reasoning in large language models," Proceedings of the 36th Conference on Neural Information Processing Systems (NeurIPS), DOI / Identifier: ACM DL: 10.5555/3600270.3602070	Shows chain-of-thought prompting significantly improves reasoning by encouraging intermediate steps. Landmark paper linking output structure and reasoning stability to reliability. Relevant because chain-of-thought and self-consistency both suggest stable reasoning traces support more reliable outputs.	• Applied structured step-by-step prompts to large-scale LLMs. • Evaluated on arithmetic, commonsense, and symbolic reasoning benchmarks. • Showed large gains from prompting alone without fine-tuning. • Links output structure to reasoning reliability.
A. J. Thirunavukarasu, D. S. J. Ting, K. Elangovan, L. Gutierrez, T. F. Tan, and D. S. W. Ting, 2023, "Large language models in medicine," Nature Medicine, 29(8), pp. 1930–1940, DOI: 10.1038/s41591-023-02448-8.	Broad review of LLM applications in medicine including diagnosis support, clinical text generation, patient communication, and drug discovery. Highlights key risks: hallucination, bias, and mismatch between benchmark success and	• Systematic review of LLM medical applications across specialties. • Assessed hallucination, bias, and reliability risks in clinical settings. • Highlighted gap between benchmark performance

	real-world trustworthiness. Important framing paper for the medical relevance of semantic stability evaluation.	and real-world trustworthiness. • Provides domain-level framing for the importance of reliability in medicine.
W. Boag, D. Doss, T. Naumann, and P. Szolovits, 2018, "What's in a note? Unpacking predictive value in clinical note representations," AMIA Joint Summits on Translational Science Proceedings, DOI / Identifier: PMC5961801.	Studies how different representations of clinical notes affect predictive performance. Shows unstructured text contains important information that structured codes alone cannot replace. Background reference reinforcing the importance of meaning-rich clinical text and the need to evaluate how models handle it consistently.	• Analysed multiple clinical note representation strategies. • Compared unstructured text against structured code-based representations. • Evaluated on clinical outcome prediction tasks. • Showed unstructured text carries information that codes do not capture.
A. Névéol, H. Dalianis, S. Velupillai, G. Savova, and P. Zweigenbaum, 2018, "Clinical natural language processing in languages other than English: Opportunities and challenges," Journal of Biomedical Informatics, DOI: 10.1186/s13326-018-0179-8	Reviews the state of clinical NLP beyond English, showing major gaps in corpora, annotation standards, and model availability. Most semantic consistency and biomedical LLM research are English-centred an important scope limitation explicitly acknowledged in the present study.	• Reviewed clinical NLP systems and corpora in 12+ non-English languages. • Identified gaps in annotation standards and model availability. • Showed the field is heavily English-focused. • Directly motivates the scope limitation acknowledged in the present study.
F. Zhu, W. Lei, C. Wang, J. Zheng, S. Poria, and T.-S. Chua, 2021, "Retrieving and reading: A comprehensive survey on open-domain question answering," arXiv preprint, DOI / Identifier: arXiv:2101.00774	Reviews open-domain QA systems including retrieval, reading comprehension, and generation approaches. Covers major QA benchmarks: SQuAD, TriviaQA, Natural Questions. Provides broader context for why	• Comprehensive survey of open-domain QA system architectures. • Covered retrieval-based, reading comprehension, and generation approaches.

	QA tasks are commonly used to study semantic uncertainty and model understanding across domains.	• Reviewed SQuAD, TriviaQA, and Natural Questions benchmarks. • Provides background for using SQuAD 2.0 as cross-domain validation.
6. Ethical Considerations Identified in Reviewed Papers		
Ethical Issue	**How Addressed in Reviewed Papers**	**Relevance to Present Study**
Patient data privacy	S. Mohan and D. Li (2019), N. J. Dobbins (2025), P. Hager et al. (2024), and K. Singhal et al. (2023) use de-identified or publicly released datasets. MedMentions uses PubMed abstracts (public domain). BioASQ releases publicly available QA pairs with no patient identifiable data.	The present study uses only publicly available de-identified datasets (MedMentions, SQuAD 2.0, BioASQ). No patient data or private clinical records are accessed at any stage.
Hallucination and clinical safety risk	P. Hager et al. (2024) and M. Agrawal et al. (2025) explicitly warn that LLM outputs can be clinically unsafe. Z. Ji et al. (2023) and L. Huang et al. (2025) show fabricated outputs are a persistent risk. Strong benchmark performance does not guarantee safety (K. Singhal et al., 2023).	The present study directly addresses this risk by providing a reliability metric beyond accuracy. No model outputs are used for clinical decision-making, diagnosis, or patient care under any circumstance.
Reproducibility and transparency	Kuhn et al. (2023) and Farquhar et al. (2024) release code and evaluation datasets. J. Novikova et al. (2025) highlight poor reproducibility in consistency research. Pre-registration is	All confirmatory analyses are pre-registered on OSF before data collection. Full code, perturbation datasets, model outputs, and entropy scores are released publicly under MIT licence.

	recommended as standard practice for confirmatory research.	
English-centric scope	A. Nevéol et al. (2018) and J. Novikova et al. (2025) highlight that most clinical NLP and consistency research is English-only, creating a systematic bias in the evidence base.	The present study is scoped to English-language clinical text. This limitation is explicitly acknowledged as a scope condition in the research questions and the discussion section.
Responsible LLM use in research	M. Sclar et al. (2024) and S. Kadavath et al. (2022) warn against over-trusting LLM outputs in research and evaluation settings. H. Raj et al. (2023) discuss risks of using consistency metrics without rigorous validation.	All LLM inference is conducted in a controlled research environment. Models are objects of analysis only, not deployed tools. No participant data is involved, and no clinical advice is generated.